

JAMES AFFUL	
+233 24 296 9042 james.afful47@gmail.com GitHub: jamesafful25 DevOps & Software Engineer Data Scientist Economist Corporate Finance Specialist	
EXECUTIVE SUMMARY	
- Talented IT, finance, investment and economics professional with over thirteen years' combined experience in software development, devOps engineering, diagnostics analysis of an economy, investment banking, financial and data analysis, risk management, research, credit appraisal, corporate and project finance, equity and fixed income trading, banking and accounting. - An experienced DevOps engineer with a knack for implementing continuous integration and delivery pipelines that have increased deployment frequency by 40%. Significantly reduced company-wide system outage time by 15% through proactive infrastructure monitoring. Worked across various industries, from tech startups to established corporations, bringing a diverse understanding of DevOps challenges and solutions. - An experienced Software Engineer (full stack) with hands on experience in designing, coding, and testing software applications. Proficient in various programming languages including JavaScript, Python, Node.js, Express.js, Java and React.js. - An experienced Data Scientist proficient in Python programming, machine learning, statistical analysis, data exploratory and deep learning with a lot of working experience in the development /building of predictive models. - An experienced Data Analyst/Business Intelligence specialist proficient in Python programming, SQL, Excel, Power BI, Tableau, Qlik Sense and QlikView with vast experience in database management, spreadsheet manipulation and dashboard creation, ensuring informed decision-making through insightful visualizations. - Expert (hands on experience) in financial modelling for Manufacturing, Agriculture, Service (financial institutions, hospitality/hotels, education, ICT, Health etc.), Infrastructure (PPP projects, road, housing, energy), Natural Resources (mining, oil & gas, renewable energy etc.) projects/companies, Leverage Buyout and Mergers and Acquisition.	
SPECIAL SKILLS	
Cloud Computing: AWS, AZURE Operating System: Linux, Windows Languages: Python, JavaScript (Node.js, Express.js and React.js) and Bash, Version Control: Git, GitHub CI-CD: GitHub Actions, Jenkins, CircleCI Container & Orchestration: Docker, Kubernetes Monitoring & Observability: Prometheus, Grafana, Xabix Infrastructure-as-Code & Configuration Mgt.: Terraform, Cloud formation, Ansible Database Management & Manipulation: MySQL, MongoDB, PostgreSQL. Data Analysis & Visualization: Machine Learning, Financial modelling, Power BI ,Tableau, Excel, STATA, Eviews and SPSS	
EXPERIENCE	
JOMACS IT SOLUTIONS	Calgary, Canada
Senior DevOps Engineer	October 2021 – Present
- Developed and implemented CI/CD pipelines using Docker, terraform etc that resulted in a 65% reduction in deployment time and increased software quality by 90%. - Directed a team of 10+ DevOps engineers, QA, and operations teams to drive initiatives and continuous improvement with a 70% success rate. - Collaborated with 10+ security teams to implement and maintain secure coding practices, vulnerability scanning and compliance standards and received 100% positive feedback. - Created and executed deployment automation procedures using Ansible and Terraform which helped to minimize errors by 80% and raised release frequency by 89%. - Designed and implemented a secure cloud infrastructure that resulted in a 40% decrease in security incidents. - Implemented a cloud based data storage system that improved data retrieval times by 45% and significantly increased the efficiency of workflows across various departments. - Successfully migrated 100+ on-premise applications to the cloud using Terraform and Docker, improving system reliability by 30% and reducing operating costs by 15%. - Designed and implemented a secure cloud infrastructure with monitoring using Prometheus and Grafana which resulted in a 40% decrease in security incidents and faster incident response.	
PHRONTLYNE TECHNOLOGIES LTD	Accra, Ghana
Software Engineer and Lead Data Scientist	February 2017 – October 2021

- Led a team that successfully developed and launched a new software application, which resulted in a revenue increase of \$1 million within the first six months of its release.
- Designed and implemented a micro services-based backend for a new financial analytics platform, resulting in a 30% increase in processing efficiency.
- Developed and deployed a comprehensive insurance application for some insurance companies in Ghana, enhancing policy management, claims processing, and customer engagement. Collaborated with cross-functional teams to design and implement robust system architecture, ensuring seamless integration with existing platforms. Optimized performance and user experience, resulting in faster claims processing and improved client satisfaction. Spearheaded the development of secure data handling protocols, ensuring compliance with industry regulations and improving data accuracy across the system.
- Developed an innovative and efficient code for a company app that increased user engagement by 35% and led a project team of seven to achieve software enhancements that improved server response time by 40%. Also, implemented proactive monitoring and logging solutions, enabling faster issue detection and reducing debugging time by 40%.
- Managed a critical update of an existing software product, resulting in a 25% improvement in performance and contributing to a 15% increase in sales for the fiscal year.
- Designed and deployed a machine-learning algorithm with 87% accuracy for predicting loan eligibility for a bank. Usage of the model reduced credit appraisal turnaround time and improved the loan portfolio quality by approximately 50%.
- Developed and deployed advanced predictive models to forecast the price of products and sales of companies (retail, real estate and manufacturing companies) using machine learning algorithms with about 85 % prediction accuracy resulting in about 35% increase in sales for the companies.
- Developed a predictive analytics model to forecast market trends, achieving a 15% reduction in forecasting errors. Also designed and implemented a real-time data visualization dashboard that monitored key performance indicators for stakeholder reporting. Also authored a company-wide data governance framework, which streamlined data collection standards and enhanced data quality.
- Developed and implemented a comprehensive business intelligence strategy for a manufacturing and distribution/retail company that resulted in a 20% increase in revenue within the first year. By analyzing market trends and customer behavior, I was able to identify new growth opportunities and optimize existing processes. Additionally, I led a team of data analysts, fostering a collaborative and results-driven environment that consistently exceeded expectations.
- Created dashboards and reports using Power BI, Tableau and Excel to communicate insights effectively to numerous stakeholders of companies. This helped the companies to make informed decision for their operations.
- Created and implemented standards, policies, and procedures for the business intelligence activities to ensure a common look and feel across reports, dashboards, and tools for numerous companies. Also created complex SQL queries to investigate data or to troubleshoot business objects queries for companies.

CDH SECURITIES LTD

Accra, Ghana

Manager, Corporate Finance and Research/Insight

January 2015 – February 2017

- Conducted a diagnostic assessment/analysis of the Ghanaian economy (private sector diagnostic analysis) to identify gaps (business opportunities and challenges) in the various sectors. This helped the holding company and other private investors to set up businesses in the various sectors of Ghana. Also provided advisory services, developed innovative financial products/services, and business plan/proposals for some companies. These helped the companies to raise capital (debt & equity) from funding agencies and banks.
- Executed valuations for some companies using various models, including cash based method/approach (discounted cash flows method, dividend discount model, residual income model, leveraged buyout) market based method (precedent transactions, relative peer multiples), Assets based model and liquidation model and also prepared prospectus for the listing of these companies on the Ghana Stock Exchange. This enabled the companies to raise funds in the form of equity from the public to support their operations. These transactions also contributed to growing the group consolidated net income by 30%.
- Executed valuations and conducted due diligence on some companies in Ghana to ascertain the fair market values of these companies. This enabled the holding company and other investors to purchase a stake (private equity financing) in these companies; thereby contributing to growing the group consolidated net income by 45%. Also, participated in the conduct of feasibility studies and prepared business plan/proposal to raise US\$8million funds to establish a five-star hotel for the holding company, thereby contributing to growing the group consolidated net income by 15%.

- Managed the holding company's private equity investment portfolio. I was responsible for monitoring, identification, resolving of potential issues on the investment/projects, writing of quarterly status report, and making recommendations to management. - Conducted a research on the effect of macroeconomic variables notably inflation, exchange rate, interest rate and fiscal instruments notably taxes and government spending on foreign direct investment and standard of living in the Ghanaian economy for a client (funding agency). The findings from the analysis helped the funding agency to make decision on their investments in Ghana. - Conducted daily, weekly, monthly, quarterly, mid-year and annual research on the performance of stocks, bonds, money market instruments and macroeconomics variables in the Ghanaian economy, Africa and the Global economy to feed the company with information on the performance of these instruments as well as the general economy. This helps the company to take strategic decisions on investments, also keep our customers abreast with the happening in the domestic and global economy, and also help them to make good investment decision. - Data collection and analysis using mathematical models and statistical techniques, forecasting of key macroeconomic variables, identification of potential risk in the economy and their likely effect on the performance of investments portfolio thereby helping the company's customers to take good investment decision to generate good return on their investments.	
CDH SAVINGS & LOANS Co. LTD (A subsidiary of CDH Investments Holdings)	Accra, Ghana
Senior Credit Risk Analyst	March 2013 – January 2015
- Created credit analytics models and risk assessment tools for the bank, usage of which reduced credit appraisal turnaround time by about 70% and increased loan portfolio size by approximately 60% over the last two years. - Managed the bank's loan portfolio of about US\$ 120 million and initiated end of day portfolio reviews and peer benchmarking to proactively manage portfolio risks, which contributed to reducing non-performing loans by 45%. - Upgraded credit portfolio management guidelines and prepared credit risk manuals and trained staff on application. These significantly enhanced skills in credit administration bank-wide and ultimately reduced defaults and improved asset quality. - Prepared routine weekly reports to guide management's review of portfolio dynamics, risks and exposure concentration. This triggered key restructuring transactions and other managerial measures that improved the bank's portfolio quality by about 40%. - Investigated validity and robustness of collaterals underlying the bank's loans. Also worked with the legal department to prepare and perfect security packages for the bank's loan portfolio, ensuring alternative recovery options in case of defaults - Structured GHC25million long-term funding solution for a Public- Private Partnership (PPP) project. Thus, the establishment of a transport terminal and shops by CDH S&L and a Municipal Assembly through build operate and transfer (BOT) agreement. The project was one of the bank's best performing investments.	
PAN AFRICAN SAVINGS AND LOANS CO. LTD (A subsidiary of Ecobank Group)	Accra, Ghana
Credit Analyst	June 2008 – March 2013
- Structured GHC4 million medium-term funding solution to the Ghana Private Road Transport Union(GPRTU) (Nsawam and Circle branches) to purchase new vehicles/buses for their operations. The project was one of the bank's best performing investments. - Structured GHC3 million medium-term funding solution to an important client (company) who was into the manufacturing of detergents and liquid soap. This helped the company to expand it operation, became more efficient and profitable. The project was one of the bank's best performing investments. - Selected to work directly with Head of Product Development to conduct quarterly market research that led the bank to develop additional loan and investment products. This diversified the bank's offerings and grew clientele by 30% and net income by 43%. - Conducted analysis of clients'/borrowers businesses to determine the viability of such businesses and then recommended to credit committee whether loan request by the borrowers must be granted or decline. - Monitored and managed loan portfolio of about GHC 10 million for two years without arrears. This was achieved through routine monitoring, end of day portfolio reviews and proactive identification of risk exposure and quick mitigation. - Maintained healthy business and good customer relationships with clients, which led to 100% retention of clients. This helped the bank to earn steady income annually and saved on advertisement and business promotions costs.	
BARLAYS BANK GHANA LTD	Accra, Ghana
Lead Generator - Sales and Customer Relationship Management	August 2007 – June 2008

- Worked with senior sales manager to perform industry analysis, risk assessment and financial due diligence on companies for the signing of scheme loan contracts worth about GH¢1 million with key medium to large-sized companies in Ghana, thereby growing the bank's income stream by about 5% - Marketed the bank's fixed deposit products to leading corporate organizations and executed and closed investment deals aggregating about GH¢ 20 million with these institutions for the bank. This increased the deposit base of the bank significantly.	
EDUCATION	
KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY	Kumasi, Ghana
<i>Master of Science (M.Sc) - Economics</i>	2017 -2018
SIKKIM MANIPAL UNIVERSITY OF HEALTH, MEDICAL AND TECHNOLOGICAL SCIENCES, INDIA (Ghana Campus)	Accra, Ghana
<i>Master of Business Administration (MBA)- Finance</i>	2009 – 2011
UNIVERSITY OF CAPE COAST	Cape Coast, Ghana
<i>Bachelor of Education (B.Ed) Psychology – Economics and Management (2nd class honors)</i>	2002 – 2006
ADDITIONAL INFORMATION	
Professional certifications: - Data Science (Python, Machine learning, Statistical Data Analysis, Exploratory Data Analysis, Data Visualization and Reporting etc)- Ghana-India Kofi Annan Centre of Excellence in ICT - Certified Loans and Risk Management Professional (Senior Member of the Institute of Loans and Risk Management, Ghana) Other skills: - Certificate in MS Word, MS Excel, MS PowerPoint – Computer Center, University of Cape Coast, Ghana - Proficiency in practical budgeting, forecasting, risk analysis and cost control using Excel – Innovative Training Solutions Institute, South Africa.	
RESEARCH CONDUCTED	
- The nexus between inflation and stock market performance in Ghana.,2023. - Effect of taxes, lavies and exchange rate volatility on ex-pump price of petrol in Ghana, 2022 - Effect of monetary policy rate on food and non-food inflations in Ghana,2020 - Effect of covid -19 pandemic on stock market performance in Ghana, 2020 - Impact of Petroleum Price Fluctuations on Inflation in Ghana, 2019 - Impact of Money Supply on Inflation in Ghana., 2019 - Impact of Foreign Direct Investment(FDI) on GDP in Ghana, 2018 - Causes of Loan default in Micro Finance Institution in Ghana, 2011 - Assessment of attitude of teachers towards the implementation of basic education reform in Ghana, 2006 - Impact of Money Supply on Interest Rate in Ghana,2005	